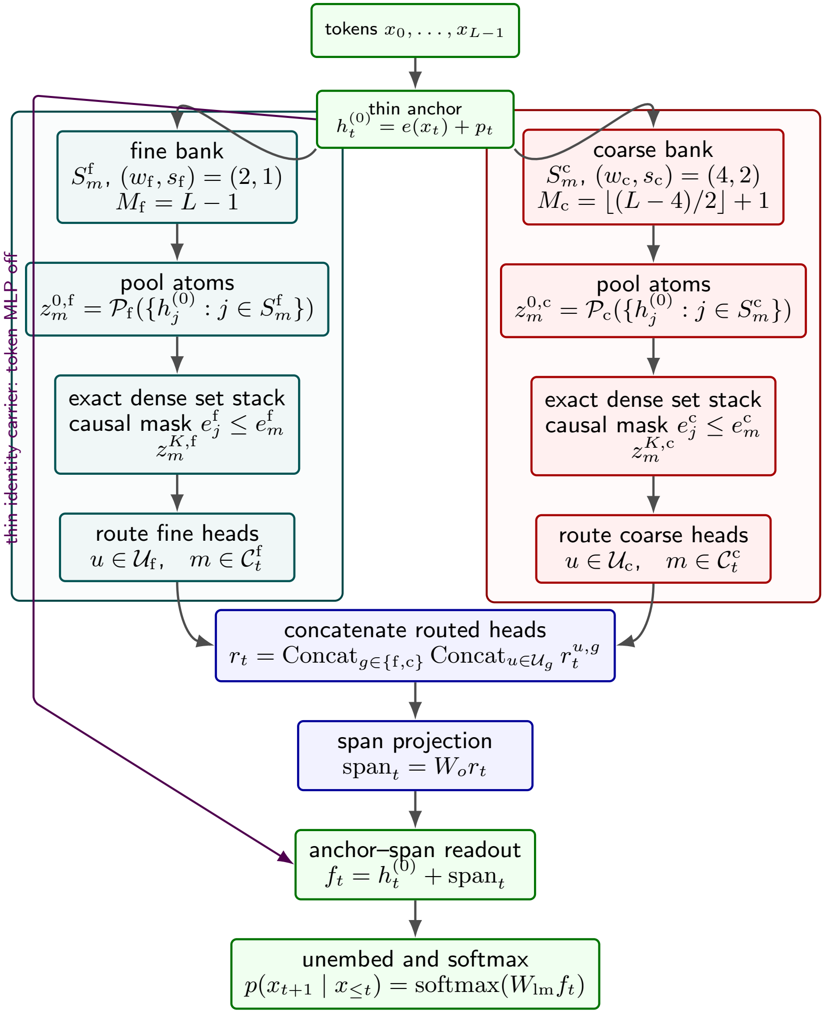
Causal multiresolution set-dictionary attention



Aligned computation. For $g \in \mathcal{G} = \{\mathrm{f}, \mathrm{c}\}$:

$$
\begin{aligned}
&\text{anchor:} && h_t^{(0)} = e(x_t) + p_t, \\
&\text{bank:} && S_m^g = \{a_m^g, \ldots, e_m^g\}, \quad e_m^g = m s_g + w_g - 1, \\
&\text{fiber:} && \mathcal{C}_t^g = \{m : \ t - w_g < e_m^g \le t\}, \\
&\text{pool:} && z_m^{0,g} = \mathcal{P}_g(\{h_j^{(0)} : j \in S_m^g\}), \\
&\text{set stack:} && z^{K,g} = \Phi_g(z^{0,g};\ e_j^g \le e_m^g), \\
&\text{route:} && \pi_{t,m}^{u,g} = \operatorname*{softmax}_{m \in \mathcal{C}_t^g}(\tilde{g}_{t,m}^{u,g}/\tau_{r,\mathrm{eff}}), \\
& && r_t^{u,g} = \sum_{m \in \mathcal{C}_t^g} \pi_{t,m}^{u,g}\, \mathrm{slice}_{u,g}(z_m^{K,g}), \\
&\text{merge:} && r_t = \mathrm{Concat}_g\, \mathrm{Concat}_{u \in \mathcal{U}_g}\, r_t^{u,g}, \\
&\text{span:} && \mathrm{span}_t = W_o r_t, \\
&\text{decode:} && f_t = h_t^{(0)} + \mathrm{span}_t, \quad \ell_t = W_{\mathrm{lm}} f_t.
\end{aligned}
$$

Current branch guards.

$$
\begin{gathered}
\mathcal{U}_{\mathrm{f}} \sqcup \mathcal{U}_{\mathrm{c}} = \{1, \ldots, H\}, \qquad H_{\mathrm{f}} + H_{\mathrm{c}} = H = 8, \\
(w_{\mathrm{f}}, s_{\mathrm{f}}) = (2,1), \qquad (w_{\mathrm{c}}, s_{\mathrm{c}}) = (4,2), \\
\texttt{anchor\_span}, \qquad \text{token MLP off}, \\
\text{trained anchor off}, \qquad \text{exact dense set backend.}
\end{gathered}
$$

The anchor rail carries current-token identity only. Historical context reaches the logits through the routed span $W_o r_t$, which is the path analyzed in Appendix B.